

CODE-GRAPH — KZTEK Labeling Studio

Tài liệu bản đồ codebase. Mọi coding agent PHẢI đọc file này TRƯỚC khi mở source.
Cập nhật ngay sau mỗi PR merge có thay đổi cấu trúc/API/schema.

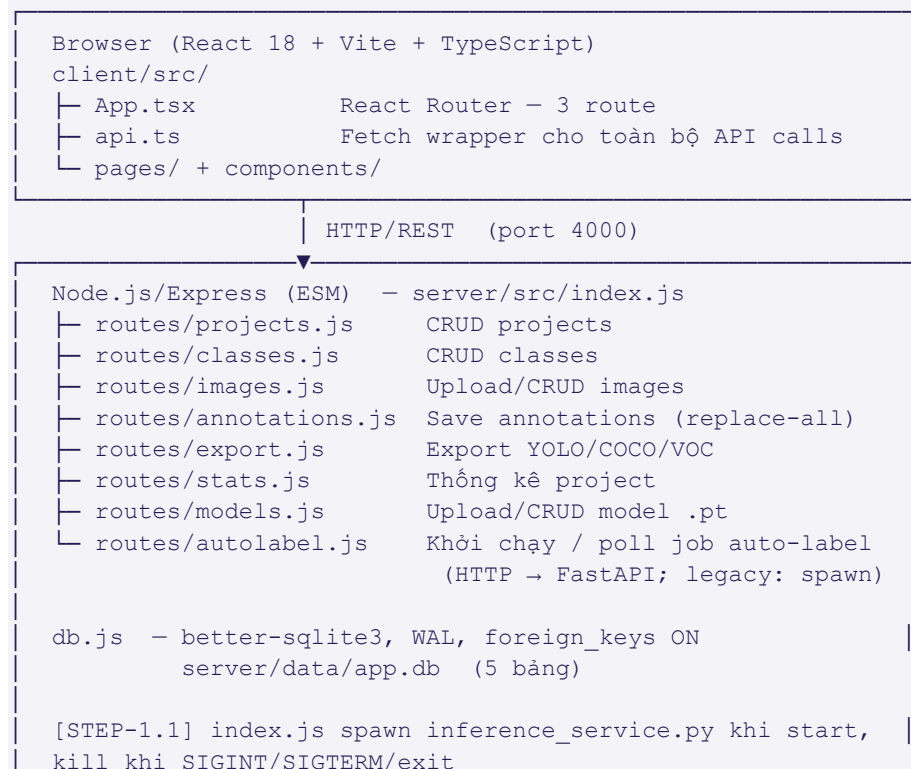
>

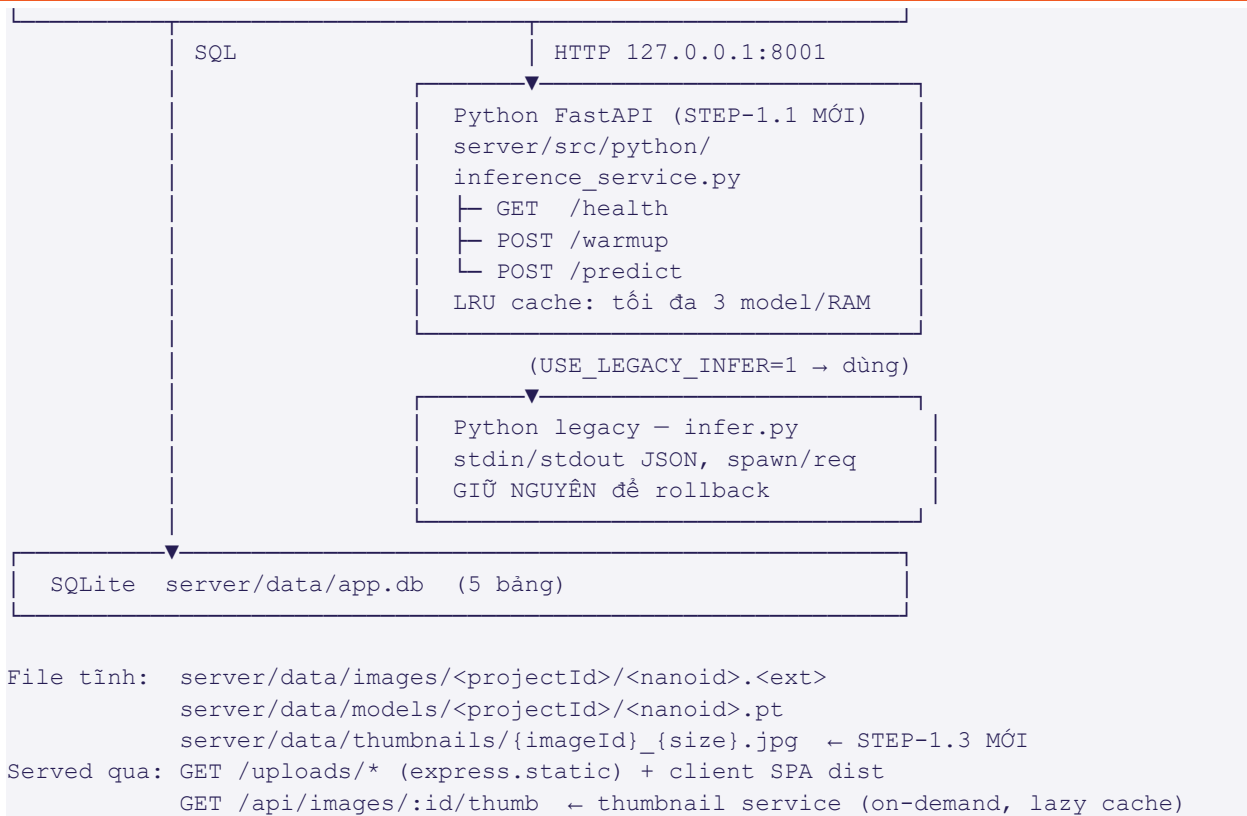
Last verified: 2026-08-04 | Cập nhật: junior-developer (STEP-1.3)

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1. Tổng quan kiến trúc





Port: 4000 (server + serve SPA dist + /uploads)

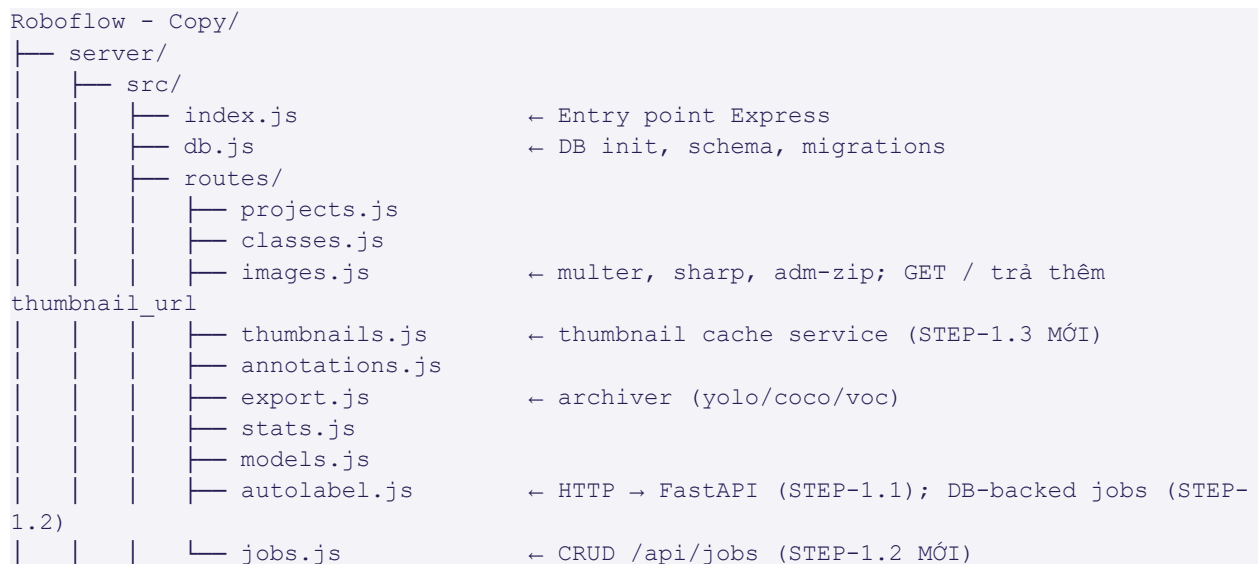
Inference port: 8001 (env `INFERENCE_PORT`, mặc định 8001)

DB file: `server/data/app.db` (tạo tự động nếu chưa có)

Python bin: env `PYTHON_BIN` hoặc `python` (mặc định)

Rollback: env `USE_LEGACY_INFER=1` → dùng `infer.py` spawn-per-request

2. Cấu trúc thư mục



```

├── python/
│   ├── inference_service.py ← FastAPI thường trực, LRU cache 3 model (STEP-
1.1 MỚI)
│   └── infer.py ← stdin→stdout JSON batch inference (GIỮ LẠI,
rollback)
├── migrations/ ← SQL migration files (chỉ tham khảo; db.js inline
là nguồn sự thật)
│   └── 001_create_jobs.sql ← (STEP-1.2)
├── data/ ← RUNTIME (git-ignored)
│   ├── app.db
│   ├── images/<projectId>/
│   ├── thumbnails/ ← cache thumbnail (STEP-1.3 MỚI):
│   │   └── {imageId}_{size}.jpg
│   └── models/<projectId>/
├── package.json
├── client/
│   ├── src/
│   │   ├── main.tsx
│   │   ├── App.tsx ← React Router setup
│   │   ├── api.ts ← Tất cả fetch call
│   │   ├── types.ts ← TS interfaces
│   │   ├── styles.css
│   │   └── pages/
│   │       ├── ProjectsPage.tsx ← List/create/delete projects
│   │       ├── ProjectDetailPage.tsx ← Image grid, upload, filters
│   │       └── AnnotatorPage.tsx ← Canvas editor (bbox+quad), autosave
│   ├── components/
│   │   ├── Logo.tsx
│   │   ├── StatsPanel.tsx
│   │   ├── ExportModal.tsx
│   │   └── AutoLabelModal.tsx
│   ├── utils/
│   │   └── files.ts
│   └── package.json
├── code-graph/
│   ├── CODE-GRAPH.md ← file này
│   └── CODE-GRAPH.pdf
└── docs/plans/ ← Plan files
  
```

3. Module map — Server

3.1 `server/src/index.js` — Entry point

Thuộc tính	Giá trị	Confidence		
---	---	---		
Loại	Express app entry	CONFIRMED		
Port	`process.env.PORT` \	\	4000`	CONFIRMED
Middleware	<code>cors()</code> , <code>express.json(limit 10mb)</code> , <code>express.static(/uploads)</code>	CONFIRMED		
SPA fallback	serve <code>client/dist/index.html</code> cho mọi non-API route	CONFIRMED		
Callers/Used-by	(entry point — không ai import)	CONFIRMED		
Imports	db.js (UPLOAD_DIR), tất cả 8 route files,	CONFIRMED		

	node:child_process (spawn)			
[STEP-1.1] Inference lifecycle	startInferenceService() → spawn inference_service.py sau app.listen(). stopInferenceService() đăng ký qua process.on('exit'/'SIGINT'/'SIGTERM')	CONFIRMED		
[STEP-1.2] Jobs router	import + mount routes/jobs.js tại /api/jobs	CONFIRMED		
[STEP-1.3] Thumbnails router	import + mount routes/thumbnails.js tại /api/images	CONFIRMED		
Env vars đọc	PORT, PYTHON_BIN, INFERENCE_PORT, USE_LEGACY_INFER	CONFIRMED		
Last verified	2026-08-04 (STEP-1.3)	-		

Route mounting:

/api/projects	→ routes/projects.js	
/api/projects/:projectId/classes	→ routes/classes.js	
/api/projects/:projectId/images	→ routes/images.js	
/api/images/:imageId/annotations	→ routes/annotations.js	
/api/projects/:projectId/export	→ routes/export.js	
/api/projects/:projectId/stats	→ routes/stats.js	
/api/projects/:projectId/models	→ routes/models.js	
/api/projects/:projectId/auto-label	→ routes/autolabel.js	
/api/jobs	→ routes/jobs.js	[STEP-1.2 MỚI]
/api/images	→ routes/thumbnails.js	[STEP-1.3 MỚI]

3.2 `server/src/db.js` — Database layer

Thuộc tính	Giá trị	Confidence
---	---	---
Loại	Singleton DB module (ESM export)	CONFIRMED
Exports	db (Database instance), DATA_DIR, UPLOAD_DIR, MODEL_DIR	CONFIRMED
Callers/Used-by	index.js (UPLOAD_DIR), projects.js, classes.js, images.js, annotations.js, export.js, stats.js, models.js, autolabel.js, jobs.js	CONFIRMED
DB engine	better-sqlite3, WAL mode, foreign_keys=ON	CONFIRMED
Schema init	CREATE TABLE IF NOT EXISTS — idempotent	CONFIRMED
Migrations inline	PRAGMA table_info + ALTER TABLE cho cột thiếu	CONFIRMED
[STEP-1.2] Startup cleanup	UPDATE jobs SET status='error' WHERE status IN ('running', 'pending') — mọi job dở dang khi restart đều bị đánh dấu error	CONFIRMED
Last verified	2026-08-04 (STEP-1.2)	-

Inline migrations hiện có (chạy khi server khởi động):

1. `annotations.type` — nếu chưa có → ADD COLUMN type TEXT DEFAULT 'bbox'
2. `annotations.points` — nếu chưa có → ADD COLUMN points TEXT
3. `classes.hotkey` — nếu chưa có → ADD COLUMN hotkey TEXT
4. **[STEP-1.2]** Cleanup jobs — UPDATE running/pending → error (startup idempotent)

3.3 `server/src/routes/projects.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router() (không mergeParams)	CONFIRMED
Imports	db	CONFIRMED
Callers/Used-by	index.js (mounted /api/projects)	CONFIRMED
Last verified	2026-08-04	-

Method	Path	File:Line	Mô tả
---	---	---	---
GET	/	projects.js:9	List all projects (JOIN count images/labeled/classes)
POST	/	projects.js:20	Create project (auto-create 2 default classes)
GET	/:id	projects.js:33	Get project by id
PATCH	/:id	projects.js:39	Update name/description
DELETE	/:id	projects.js:48	Delete project (CASCADE images/classes/annotations)

3.4 `server/src/routes/classes.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db	CONFIRMED
Callers/Used-by	index.js (mounted /api/projects/:projectId/classes)	CONFIRMED
Last verified	2026-08-04	-

Method	Path	File:Line	Mô tả
---	---	---	---
GET	/	classes.js:13	List classes of project (ORDER BY sort_order)
POST	/	classes.js:19	Create class (auto sort_order)
PATCH	/:classId	classes.js:30	Update name/color/hotkey
DELETE	/:classId	classes.js:45	Delete class (CASCADE annotations)

Lưu ý: `hotkey` bị truncate còn 1 ký tự (`normalizeHotkey` function:L7).

3.5 `server/src/routes/images.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, UPLOAD_DIR, multer, sharp, adm-zip, nanoid	CONFIRMED
Callers/Used-by	index.js (mounted /api/projects/:projectId/images)	CONFIRMED
Max file size	100MB / ảnh, 2GB / zip, 20000 ảnh / upload	CONFIRMED
Định dạng	JPG, PNG, WEBP, BMP	CONFIRMED
Last verified	2026-08-04	-

Method	Path	File:Line	Mô tả
---	---	---	---
GET	/	images.js:51	List images (kèm class_ids array + thumbnail_url — STEP-1.3)
GET	/:imageId	images.js:64	Get single image + annotations
POST	/upload	images.js:73	Upload nhiều file ảnh (multer.array)
POST	/upload-zip	images.js:95	Upload ZIP chứa ảnh
PATCH	/:imageId	images.js:143	Update split/status
DELETE	/:imageId	images.js:153	Delete image + xóa file vật lý

3.6 `server/src/routes/annotations.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, nanoid	CONFIRMED
Callers/Used-by	index.js (mounted /api/images/:imageId/annotations)	CONFIRMED
Pattern	Replace-all: DELETE + bulk INSERT trong transaction	CONFIRMED
Last verified	2026-08-04	-

Method	Path	File:Line	Mô tả
---	---	---	---
PUT	/	annotations.js:12	Replace toàn bộ annotations của 1 ảnh; auto-update

		images.status
--	--	---------------

Quan trọng: Không có GET riêng — annotations được trả cùng `GET /images/:imageId` (images.js:64).

3.7 `server/src/routes/export.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, UPLOAD_DIR, archiver	CONFIRMED
Callers/Used-by	index.js (mounted /api/projects/:projectId/export)	CONFIRMED
Last verified	2026-08-04	-

Method	Path	File:Line	Mô tả			
---	---	---	---			
GET	/split-preview	export.js:85	Preview phân chia train/valid/test trước export			
GET	/	export.js:114	Export ZIP (format=yolo\	coco\	voc, splitMode=manual\	auto)

Hỗ trợ: YOLO (bbox + OBB quad), COCO JSON, Pascal VOC XML. Stratified split logic (rarest class first).

3.8 `server/src/routes/stats.js`

Method	Path	File:Line	Mô tả
---	---	---	---
GET	/	stats.js:6	Tổng số ảnh, labeled, bySplit, perClass annotation count

3.9 `server/src/routes/models.js`

Method	Path	File:Line	Mô tả
---	---	---	---
GET	/	models.js:26	List models của project
POST	/upload	models.js:32	Upload file .pt (max 1GB)
DELETE	/:modelId	models.js:43	Delete model + xóa file .pt

3.10 `server/src/routes/autolabel.js`

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, UPLOAD_DIR, MODEL_DIR, child_process.spawn, readline, node:path, nanoid	CONFIRMED
[STEP-1.2] Job storage	DB-backed: <code>createJobInDB()</code> khi POST; <code>syncJobToDB()</code> khi hoàn thành; <code>getJobFromDB()</code> khi GET fall-back. In-memory Map chỉ dùng trong quá trình inference đang chạy.	CONFIRMED
[STEP-1.1] Inference mode mặc định	HTTP fetch → <code>http://127.0.0.1:{INFERENCING_PORT}/predict</code> (FastAPI service thường trực)	CONFIRMED
[STEP-1.1] Rollback mode	<code>USE_LEGACY_INFER=1</code> → dùng spawn infer.py stdin/stdout cũ (giữ nguyên để rollback)	CONFIRMED
Health gate	Trước mỗi job, gọi <code>GET /health</code> (timeout 3s); trả 503 nếu service chưa sẵn sàng	CONFIRMED
Batch size	32 ảnh/request tới FastAPI; timeout 5 phút/batch	CONFIRMED
Env vars đọc	<code>INFERENCING_PORT</code> (default 8001), <code>USE_LEGACY_INFER</code>	CONFIRMED
Last verified	2026-08-04 (STEP-1.2)	-

Method	Path	File:Line	Mô tả
---	---	---	---
POST	/	autolabel.js:~290	Khởi chạy job inference; tạo row DB ngay; returns { <code>jobId</code> , <code>total</code> } 202
GET	/: <code>jobId</code>	autolabel.js:~350	Poll trạng thái: memory Map → fall-back DB

Cơ chế mới (STEP-1.1): `checkInferenceHealth()` → `fetch(INFERENCING_URL/predict, batch)` → parse response {`classes`, `results`, `errors`} → `saveDetectionsForImage()` → cập nhật job.

Cơ chế persist (STEP-1.2): `createJobInDB()` khi POST; HTTP mode `.then()/ .catch()` → `syncJobToDB()` → `jobs.delete()`. Legacy mode: `setInterval(500ms)` poll job.status → sync khi done/error.

Legacy (USE_LEGACY_INFER=1): `spawn(PYTHON_BIN, [INFER_SCRIPT])` → stdin JSON → readline stdout — giữ nguyên, không xóa.

Depth-1 callers: index.js (mount `/api/projects/:pid/auto-label`) — không thay đổi mount path.

3.12 `server/src/routes/thumbnails.js` — Thumbnail cache service (STEP-1.3 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router() (không mergeParams)	CONFIRMED
Imports	db, UPLOAD_DIR, DATA_DIR, path, fs, sharp	CONFIRMED
Callers/Used-by	index.js (mounted <code>/api/images</code>)	CONFIRMED
Cache dir	<code>DATA_DIR/thumbnails/</code> — tạo sẵn khi module load	CONFIRMED

Naming cache file	{imageId}_{size}.jpg	CONFIRMED
Size range	min 32, max 600, default 300	CONFIRMED
Fallback	resize lỗi → serve ảnh gốc (không trả 500)	CONFIRMED
Last verified	2026-08-04 (STEP-1.3)	-

Method	Path	Mô tả
---	---	---
GET	/:id/thumb	Lazy thumbnail: cache hit → serve; ảnh gốc nhỏ hơn size → serve gốc; else resize+cache+serve

Query params: ?size=300 (default 300, min 32, max 600)

Cache-Control: public, max-age=86400 (1 ngày)

Exports: THUMB_DIR (dùng để test/cleanup)

3.11 `server/src/routes/jobs.js` — Job history CRUD (STEP-1.2 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router()	CONFIRMED
Imports	db	CONFIRMED
Callers/Used-by	index.js (mounted /api/jobs)	CONFIRMED
Last verified	2026-08-04 (STEP-1.2)	-

Method	Path	File:Line	Mô tả
---	---	---	---
GET	/	jobs.js:~22	List jobs by projectId (query param ?projectId=) — ORDER BY created_at DESC
GET	/:id	jobs.js:~31	Get single job by id
DELETE	/:id	jobs.js:~37	Delete job record

Response shape (`normalizeJob`): {id, projectId, status, total, done, created, failed, modelId, error, unmatchedClasses[], createdAt, updatedAt}

3.13 `server/src/routes/reviews.js` — Review workflow (STEP-2.3 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Router options	Router({ mergeParams: true })	CONFIRMED
Imports	db, requireRole (middleware/roles.js)	CONFIRMED
Callers/Used-by	index.js (mounted /api/images/:imageId)	CONFIRMED
Last verified	2026-08-04 (STEP-2.3)	-

Method	Path	Role required	Mô tả
---	---	---	---
POST	/submit-review	annotator, admin	draft\ rejected → in_review
POST	/approve	reviewer, admin	in_review → approved
POST	/reject	reviewer, admin	in_review → rejected (body: {comment})

Lifecycle: draft → in_review → approved hoặc rejected → (submit lại) → in_review. Conflict trạng thái sai → 409 REVIEW_STATUS_CONFLICT.

4. Module map — Client

4.1 `client/src/App.tsx` — Router root

Route	Component	Mô tả
---	---	---
/	ProjectsPage	Danh sách projects
/projects/:projectId	ProjectDetailPage	Chi tiết project, image grid
/projects/:projectId/annotate/:imageId	AnnotatorPage	Canvas annotator

4.2 `client/src/api.ts` — API client

Thuộc tính	Giá trị	Confidence
---	---	---
Pattern	Thin fetch wrapper (request<T>) — không có state	CONFIRMED
Callers/Used-by	ProjectsPage, ProjectDetailPage, AnnotatorPage	CONFIRMED
Auth	Không có (no token/cookie)	CONFIRMED
Last verified	2026-08-04	-

Hàm public:

```
listProjects, createProject, getProject, deleteProject
listClasses, createClass, updateClass, deleteClass
listImages, getImage, uploadImages, uploadZip, updateImage, deleteImage
saveAnnotations
exportUrl, getSplitPreview
getStats
listModels, uploadModel, deleteModel
startAutoLabel, getAutoLabelJob
```

4.3 `client/src/types.ts` — TypeScript types

Các interface chính:

- **Project** — id, name, description, created_at, image_count, labeled_count, class_count
- **ClassLabel** — id, project_id, name, color, sort_order, hotkey
- **ImageItem** — id, project_id, filename, original_name, width, height, split, status, created_at, class_ids[], **thumbnail_url?** (STEP-1.3)
- **Annotation** — id, image_id, class_id, x, y, w, h, type, points
- **ImageWithAnnotations** — ImageItem + annotations[]
- **ModelInfo** — id, project_id, filename, original_name, created_at
- **AutoLabelJob** — status, total, done, created, failed, error, unmatchedClasses[]

4.4 `client/src/pages/AnnotatorPage.tsx` — Canvas editor

Thuộc tính	Giá trị	Confidence
---	---	---
Callers/Used-by	App.tsx (route /projects/:pid/annotate/:iid)	CONFIRMED
API calls	api.listClasses, api.listImages, api.getImage, api.saveAnnotations	CONFIRMED
Autosave	debounce 600ms sau mỗi thay đổi box	CONFIRMED
Tools	bbox (draw rect), quad (4-point OBB)	CONFIRMED
Hotkeys	Arrow ← → navigate, Delete xóa box	INFERRED
State: boxes	Box[] (local) — sync lên server khi dirty	CONFIRMED
State: undo	KHÔNG có undo/redo hiện tại	CONFIRMED
Last verified	2026-08-04	-

4.5 `client/src/pages/ProjectDetailPage.tsx` — Image grid

API calls	listClasses, listImages, uploadImages, uploadZip, updateImage, deleteImage
---	---
Features	Filter by status/split/class/search, pagination (page×pageSize), drag-drop upload
Modals	StatsPanel, ExportModal, AutoLabelModal

4.6 Components

Component	Callers	Mô tả
---	---	---

Logo.tsx	App.tsx	KZTEK logo SVG inline
StatsPanel.tsx	ProjectDetailPage	Chart/bảng stats project
ExportModal.tsx	ProjectDetailPage	UI chọn format/split, trigger export
AutoLabelModal.tsx	ProjectDetailPage	Chọn model, confidence, scope; poll job progress

5. Module map — Python Inference

5.1 `server/src/python/infer.py` — Legacy (GIỮ LẠI để rollback)

Thuộc tính	Giá trị	Confidence
---	---	---
Protocol	stdin: JSON payload → stdout: streaming NDJSON (1 line/ảnh) + {"done":true}	CONFIRMED
Caller	autolabel.js runInference() (khi USE_LEGACY_INFER=1)	CONFIRMED
Dependencies	ultralytics (import lazy bên trong main())	CONFIRMED
Model load	YOLO(model_path) — load từ disk mỗi lần spawn	CONFIRMED
Output bbox	{class_index, type:"bbox", x, y, w, h} (pixel coords)	CONFIRMED
Output quad	{class_index, type:"quad", points:[{x,y}×4]} (OBB)	CONFIRMED
Error handling	Per-image try/except → {image_id, error, classes}	CONFIRMED
Status	GIỮ NGUYÊN — rollback bằng USE_LEGACY_INFER=1	CONFIRMED
Last verified	2026-08-04	-

Input format (stdin):

```
{
  "model_path": "/abs/path/model.pt",
  "conf": 0.25,
  "images": [{"id": "nanoid", "path": "/abs/path/img.jpg"}, ...]
}
```

5.2 `server/src/python/inference\_service.py` — FastAPI thường trực (STEP-1.1 MỚI)

Thuộc tính	Giá trị	Confidence
---	---	---
Framework	FastAPI + uvicorn	CONFIRMED
Port	127.0.0.1:{INFERENCE_PORT} (default 8001)	CONFIRMED

LRU cache	<code>LRUModelCache(max_size=3)</code> — tối đa 3 model giữ trong RAM	CONFIRMED
Caller	<code>index.js</code> <code>startInferenceService()</code> — spawn khi Node start	CONFIRMED
Dependencies	<code>fastapi</code> , <code>uvicorn</code> , <code>ultralytics</code> (lazy import bên trong <code>_load_model</code>)	CONFIRMED
Lifecycle	<code>index.js</code> spawn/kill; không auto- restart khi crash (STEP log <code>Service stopped</code>)	CONFIRMED
Env vars	<code>INFERENCE_PORT</code>	CONFIRMED
Last verified	2026-08-04 (STEP-1.1)	-

Endpoints:

Method	Path	Mô tả
---	---	---
GET	<code>/health</code>	Trả <code>{status, loaded_models[], uptime_s}</code>
POST	<code>/warmup</code>	Preload model vào LRU cache. Body: <code>{model_path}</code>
POST	<code>/predict</code>	Batch inference. Body: <code>{model_path, conf, iou, images:[{id,path}]}</code>

Response `/predict`:

```
{
  "classes": ["class_name_0", ...],
  "results": [{ "image_id": "nanoid", "detections":
    [{"class_index": 0, "type": "bbox", "x": ..., "y": ..., "w": ..., "h": ...}] },
  "errors": [{ "image_id": "nanoid", "error": "message" }]
}
```

Output bbox: `{class_index, type:"bbox", x, y, w, h}` — tương thích với `infer.py` cũ

Output quad (OBB): `{class_index, type:"quad", points:[{x,y}×4]}`

Error handling: Per-image try/except → gộp vào `errors[]` (không crash toàn batch)

6. DB Schema thực tế

Nguồn: `server/src/db.js` — đọc trực tiếp CREATE TABLE statements. Confidence: CONFIRMED toàn bộ.

Bảng `projects`

Cột	Kiểu	Ràng buộc
---	---	---
<code>id</code>	TEXT	PRIMARY KEY
<code>name</code>	TEXT	NOT NULL
<code>description</code>	TEXT	DEFAULT "
<code>created_at</code>	TEXT	NOT NULL DEFAULT

		datetime('now')
--	--	-----------------

Bảng `classes`

Cột	Kiểu	Ràng buộc
---	---	---
id	TEXT	PRIMARY KEY
project_id	TEXT	NOT NULL, FK→projects(id) ON DELETE CASCADE
name	TEXT	NOT NULL
color	TEXT	NOT NULL
sort_order	INTEGER	NOT NULL DEFAULT 0
hotkey	TEXT	nullable (migration nếu chưa có)

Index: `idx_classes_project` ON `classes(project_id)`

Bảng `images`

Cột	Kiểu	Ràng buộc	
---	---	---	
id	TEXT	PRIMARY KEY	
project_id	TEXT	NOT NULL, FK→projects(id) ON DELETE CASCADE	
filename	TEXT	NOT NULL (tên file trên disk, nanoid+ext)	
original_name	TEXT	NOT NULL (tên gốc khi upload)	
width	INTEGER	NOT NULL	
height	INTEGER	NOT NULL	
split	TEXT	NOT NULL DEFAULT 'train' (train/valid/test)	
status	TEXT	NOT NULL DEFAULT 'unlabeled' (unlabeled/labeled)	
created_at	TEXT	NOT NULL DEFAULT datetime('now')	
review_status	TEXT	NOT NULL DEFAULT 'draft' — STEP-2.3 MỚI (draft→in_review→approved\rejected)	
review_comment	TEXT	nullable — STEP-2.3 MỚI (lý do reject)	
reviewed_by	INTEGER	FK→users(id), nullable — STEP-2.3 MỚI	
reviewed_at	TEXT	nullable — STEP-2.3 MỚI	

Index: `idx_images_project` ON `images(project_id)`, `idx_images_review_status` ON `images(review_status)` (STEP-2.3 MỚI)

⚠ **Lưu ý cho STEP-3.5:** `submit-review` hiện cho phép bất kỳ ảnh draft/rejected nào (không check `completed_at`). Khi STEP-3.5 thêm `completed_at/completed_by`, sửa `reviews.js` để check `completed_at IS NOT NULL` trước khi cho submit-review.

Bảng `annotations`

Cột	Kiểu	Ràng buộc
---	---	---
id	TEXT	PRIMARY KEY
image_id	TEXT	NOT NULL, FK→images(id) ON DELETE CASCADE
class_id	TEXT	NOT NULL, FK→classes(id) ON DELETE CASCADE
x	REAL	NOT NULL (pixel, top-left)
y	REAL	NOT NULL (pixel, top-left)
w	REAL	NOT NULL (pixel width)
h	REAL	NOT NULL (pixel height)
type	TEXT	NOT NULL DEFAULT 'bbox' (bbox/quad) — migration nếu chưa có
points	TEXT	nullable, JSON string [[[x,y]×4]] cho type=quad — migration nếu chưa có
created_at	TEXT	NOT NULL DEFAULT datetime('now')
version	INTEGER	NOT NULL DEFAULT 0 — STEP-3.1 MỚI (optimistic locking, dùng ở STEP-3.4)

Index: `idx_annotations_image ON annotations(image_id)`

Bảng `annotation\_history` — STEP-3.1 MỚI

Cột	Kiểu	Ràng buộc
---	---	---
id	INTEGER	PRIMARY KEY AUTOINCREMENT
image_id	TEXT	NOT NULL, FK→images(id) ON DELETE CASCADE
version	INTEGER	NOT NULL
snapshot	TEXT	NOT NULL — JSON toàn bộ annotations của ảnh tại thời điểm save (SNAPSHOT, không diff — ADR AD-5)
actor_id	INTEGER	FK→users(id), nullable
created_at	TEXT	NOT NULL DEFAULT datetime('now')

Index: `idx_ann_history_image, idx_ann_history_img_ver(image_id, version DESC)`. **Retention:** tối đa 200 version/ảnh qua `pruneAnnotationHistory(imageId)` (export từ `db.js`, gọi bởi STEP-3.2).

Bảng `activity\_log` — STEP-3.1 MỚI

Cột	Kiểu	Ràng buộc
---	---	---
id	INTEGER	PRIMARY KEY AUTOINCREMENT
project_id	TEXT	NOT NULL, FK→projects(id) ON DELETE CASCADE
actor_id	INTEGER	FK→users(id), nullable

action	TEXT	NOT NULL (VD: 'upload', 'export', 'split_change')
detail	TEXT	nullable, JSON tùy ý
created_at	TEXT	NOT NULL DEFAULT datetime('now')

Index: `idx_activity_log_project`, `idx_activity_log_action`. Route ghi log: STEP-3.3 (bảng đã tạo, chưa có route ghi ở STEP-3.1).

⚠ **Lưu ý:** `actor_id` không có `ON DELETE CASCADE/SET NULL` (mặc định `NO ACTION`) — không thể hard-delete user đã có history/activity_log khi `foreign_keys=ON`. Dùng `is_active=0` (soft-delete, đã có từ STEP-2.2) thay vì xóa thật.

Bảng `models`

Cột	Kiểu	Ràng buộc
---	---	---
id	TEXT	PRIMARY KEY
project_id	TEXT	NOT NULL, FK→projects(id) ON DELETE CASCADE
filename	TEXT	NOT NULL (tên file .pt trên disk)
original_name	TEXT	NOT NULL (tên gốc khi upload)
created_at	TEXT	NOT NULL DEFAULT datetime('now')

Index: `idx_models_project` ON `models(project_id)`

Bảng `jobs` — STEP-1.2 MỚI

Cột	Kiểu	Ràng buộc
---	---	---
id	TEXT	PRIMARY KEY
project_id	TEXT	NOT NULL (no FK — không CASCADE khi xóa project, job history giữ lại)
status	TEXT	NOT NULL DEFAULT 'pending' (pending/running/done/error)
total_images	INTEGER	NOT NULL DEFAULT 0
processed	INTEGER	NOT NULL DEFAULT 0 (ảnh đã xử lý xong)
created_annotations	INTEGER	NOT NULL DEFAULT 0
failed	INTEGER	NOT NULL DEFAULT 0
model_id	TEXT	nullable
error_msg	TEXT	nullable
unmatched_classes	TEXT	NOT NULL DEFAULT '[]' (JSON array tên class không map được)
created_at	TEXT	NOT NULL DEFAULT datetime('now')
updated_at	TEXT	NOT NULL DEFAULT datetime('now')

Indexes: `idx_jobs_project` ON `jobs(project_id)`, `idx_jobs_status` ON `jobs(status)`

Startup behavior: Khi server khởi động, mọi row có `status IN ('running', 'pending')` → cập nhật thành `'error'` với `error_msg='Server restarted while job was in progress'`.

7. API Endpoints

Tất cả CONFIRMED từ source (index.js route mounting + từng route file).

Projects

Method	Path	File:Line	Response
---	---	---	---
GET	/api/projects	projects.js:9	Project[] (kèm counts)
POST	/api/projects	projects.js:20	Project (201)
GET	/api/projects/:id	projects.js:33	Project
PATCH	/api/projects/:id	projects.js:39	Project
DELETE	/api/projects/:id	projects.js:48	204

Classes

Method	Path	File:Line	Response
---	---	---	---
GET	/api/projects/:pid/classes	classes.js:13	ClassLabel[]
POST	/api/projects/:pid/classes	classes.js:19	ClassLabel (201)
PATCH	/api/projects/:pid/classes/:cid	classes.js:30	ClassLabel
DELETE	/api/projects/:pid/classes/:cid	classes.js:45	204

Images

Method	Path	File:Line	Response
---	---	---	---
GET	/api/projects/:pid/images	images.js:51	ImageItem[]
GET	/api/projects/:pid/images/:iid	images.js:64	ImageWithAnnotations
POST	/api/projects/:pid/images/upload	images.js:73	ImageItem[] (201)
POST	/api/projects/:pid/images/upload-zip	images.js:95	{created, skipped} (201)
PATCH	/api/projects/:pid/images/:iid	images.js:143	ImageItem
DELETE	/api/projects/:pid/images/:iid	images.js:153	204

Annotations

Method	Path	File:Line	Response
---	---	---	---
PUT	/api/images/:iid/annotations	annotations.js:12	Annotation[]

Export

Method	Path	File:Line	Response
---	---	---	---
GET	/api/projects/:pid/export/split-preview	export.js:85	SplitPreview JSON

GET	/api/projects/:pid/export	export.js:114	ZIP stream (attachment)
-----	---------------------------	---------------	-------------------------

Query params export: `format=yolo\|coco\|voc, splitMode>manual\|auto, trainRatio=0.8`

Stats

Method	Path	File:Line	Response
---	---	---	---
GET	/api/projects/:pid/stats	stats.js:6	ProjectStats JSON

Models

Method	Path	File:Line	Response
---	---	---	---
GET	/api/projects/:pid/models	models.js:26	ModelInfo[]
POST	/api/projects/:pid/models/upload	models.js:32	ModelInfo (201)
DELETE	/api/projects/:pid/models/:mid	models.js:43	204

Auto-label

Method	Path	File:Line	Response
---	---	---	---
POST	/api/projects/:pid/auto-label	autolabel.js:~290	{jobId, total} (202)
GET	/api/projects/:pid/auto-label/:jid	autolabel.js:~350	AutoLabelJob (memory → DB fallback)

Jobs (STEP-1.2 MỚI)

Method	Path	File:Line	Response
---	---	---	---
GET	/api/jobs?projectId=:pid	jobs.js:~22	JobRecord[]
GET	/api/jobs/:id	jobs.js:~31	JobRecord
DELETE	/api/jobs/:id	jobs.js:~37	{ok:true}

Thumbnails (STEP-1.3 MỚI)

Method	Path	File:Line	Response
---	---	---	---
GET	/api/images/:id/thumb	thumbnails.js:29	JPEG binary (thumbnail or original) — 200; 404 nếu image/file không tồn tại

Query params: `?size=300` (default 300px, min 32, max 600)

Cache-Control: `public, max-age=86400`

8. Dependency graph

Server dependencies (package.json — CONFIRMED)

Package	Version	Dùng ở
---	---	---
express	^4.21.0	index.js, tất cả routes
better-sqlite3	^9.6.0	db.js
cors	^2.8.5	index.js
multer	^1.4.5-lts.1	images.js, models.js
sharp	^0.32.6	images.js (metadata width/height, resize)
adm-zip	^0.6.0	images.js (unzip upload)
archiver	^7.0.1	export.js (tạo zip export)
nanoid	^5.0.7	projects, classes, images, annotations, models, autolabel

Client dependencies (package.json — CONFIRMED)

Package	Version	Dùng ở
---	---	---
react	^18.3.1	tất cả components/pages
react-dom	^18.3.1	main.tsx
react-router-dom	^6.26.2	App.tsx, các pages
vite	^5.4.6	dev/build tool
typescript	^5.6.2	type checking

Python dependencies

Package	Dùng ở	Confidence
---	---	---
ultralytics	infer.py, inference_service.py (YOLO model, predict — lazy import)	CONFIRMED
fastapi	inference_service.py (FastAPI app, endpoints, Pydantic schemas)	CONFIRMED
uvicorn	inference_service.py (ASGI server, entrypoint)	CONFIRMED

9. Điểm sẽ bị ảnh hưởng bởi plan cải tiến

Mục này là "blast radius map" cho 23 bước trong PLAN-MASTER.md.
Confidence: CONFIRMED từ phân tích source + plan.

Phase 1.1 — Inference service FastAPI (thường trực) — ✔ HOÀN THÀNH

Files đã thay đổi (STEP-1.1):

- `server/src/routes/autolabel.js` — Refactor: thêm `runInferenceHTTP()` gọi FastAPI, giữ `runInference()` cũ cho rollback (USE_LEGACY_INFER=1). Health gate trước mỗi job → 503 khi service down.
- `server/src/index.js` — Thêm `startInferenceService()` / `stopInferenceService()` lifecycle.
- `server/src/python/inference_service.py` — FastAPI app mới, LRU cache 3 model.
- `server/src/python/infer.py` — GIỮ NGUYÊN (rollback).
- `server/README.md` — Thêm mục "Chạy inference service", env vars, cấu trúc Python inference.

Depth-1 callers: `index.js` (mount `autolabelRouter`) — không đổi mount path.

Watch out: `client/api.ts` `startAutoLabel` + `getAutoLabelJob` không đổi (same REST interface).

Phase 1.2 — Persist bảng `jobs` — ✓ HOÀN THÀNH

Files đã thay đổi (STEP-1.2):

- `server/src/db.js` — thêm `CREATE TABLE IF NOT EXISTS jobs (...)` + 2 indexes + startup cleanup (UPDATE running/pending → error)
- `server/src/routes/autolabel.js` — thêm `createJobInDB`, `syncJobToDB`, `getJobFromDB`; POST handler tạo DB row + sync khi hoàn thành; GET handler fall-back DB
- `server/src/routes/jobs.js` — **MỚI:** GET/GET/:id/DELETE /api/jobs
- `server/src/index.js` — import + mount `jobsRouter` tại `/api/jobs`
- `server/migrations/001_create_jobs.sql` — **MỚI:** migration SQL (tham khảo)

Depth-1 callers: `client/src/api.ts` (`getAutoLabelJob`) — REST interface giữ nguyên, không cần sửa client.

Watch out: `jobs` table không có FK tới `projects` — khi xóa project, jobs không bị cascade. Thiết kế có chủ ý (giữ job history). Phase sau nếu cần cleanup → `DELETE FROM jobs WHERE project_id=?` trong `routes/projects.js` khi DELETE project.

Phase 1.3 — Thumbnail service — ✓ HOÀN THÀNH

Files đã thay đổi (STEP-1.3):

- `server/src/routes/thumbnails.js` — **MỚI:** lazy thumbnail endpoint GET `/api/images/:id/thumb`, cache `DATA_DIR/thumbnails/{id}_{size}.jpg`
- `server/src/index.js` — import + mount `thumbnailsRouter` tại `/api/images`
- `server/src/routes/images.js` — GET / thêm field `thumbnail_url` vào response
- `client/src/types.ts` — thêm `thumbnail_url?: string` vào `ImageItem`
- `client/src/pages/ProjectDetailPage.tsx` — image grid dùng `img.thumbnail_url` thay `/uploads/...`

Depth-1 callers:

- `client/src/pages/ProjectDetailPage.tsx` → dùng `img.thumbnail_url` (thêm)
- `client/src/pages/AnnotatorPage.tsx` → không thay đổi (dùng `getImage` riêng, không qua list)

Watch out: `THUMB_DIR` tạo khi module load — nếu `DATA_DIR` không ghi được sẽ crash lúc server start. Cần đảm bảo `server/data/` có write permission.

Phase 2.2 — Auth implementation — ✔ HOÀN THÀNH

Files bị ảnh hưởng:

- `server/src/db.js` — bảng `users` (INTEGER AUTOINCREMENT id, username, password_hash, display_name, role, color, created_at, is_active)
- `server/src/index.js` — mount `auth.js/users.js`, middleware auth
- Tạo mới: `server/src/routes/auth.js`, `server/src/routes/users.js`, `server/src/middleware/`, `server/src/lib/`
- `client/src/api.ts`, `client/src/types.ts`, `client/src/App.tsx`, `client/src/pages/LoginPage.tsx`
- Test: `tests/auth.test.js` — 87 passed theo ma trận AD-A5 (2 điều kiện CTO #A1/#A2 đạt)

Phase 2.3 — Review workflow — ✔ HOÀN THÀNH

Files bị ảnh hưởng:

- `server/src/db.js` — `m004_add_review_status()`: 4 cột mới trên `images` (`review_status`, `review_comment`, `reviewed_by`, `reviewed_at`) + index
- Tạo mới: `server/src/routes/reviews.js` (xem §3.13)
- `server/src/index.js` — mount `reviewsRouter` tại `/api/images/:imageId`
- Test: `tests/auth.test.js` Row 8 — 8 test case pass (submit-review, approve, reject, comment, role guard, 401)

Watch out cho STEP-3.5: `submit-review` hiện không check `completed_at` (field đó chưa tồn tại) — khi STEP-3.5 xong, cập nhật điều kiện submit trong `reviews.js`.

Phase 3.1 — DB Migration (ĐỤNG SCHEMA HIỆN CÓ) — ✔ HOÀN THÀNH

Files bị ảnh hưởng:

- `server/src/db.js` — `m005_phase3_schema()`: cột `annotations.version` DEFAULT 0, bảng `annotation_history` (snapshot), `activity_log`; auto-backup `.bak-{ts}` (giữ 5 bản) + verify row count trước/sau (throw nếu mất dữ liệu — CTO condition #1); export `pruneAnnotationHistory()`.
- Chưa đụng `server/src/routes/annotations.js` — route UPDATE version + INSERT history là việc của STEP-3.2, bước này CHỈ tạo schema.

Kết quả verify row count (thực tế đã chạy): projects 2→2, classes 4→4, images 1→1, annotations 0→0, models 0→0, jobs 0→0, users 1→1 — không mất dữ liệu ở bảng nào.

Depth-1 callers sẽ break nếu STEP-3.2 sai: tất cả code đọc/ghi bảng `annotations` (`annotations.js`, `images.js:64`, `export.js`, `stats.js`, `autolabel.js`) — CHƯA đổi ở bước này, cần re-verify khi STEP-3.2 sửa `annotations.js`.

Watch out: `annotation_history.snapshot` là SNAPSHOT JSON (không diff) — STEP-3.2 revert = ghi đè thẳng, không merge. `actor_id` không có ON DELETE CASCADE — không hard-delete user có history.

Phase 3.5 — Image done status

Files bị ảnh hưởng:

- `server/src/db.js` — ALTER TABLE `images` ADD COLUMN `completed_at` DATETIME, ADD COLUMN `completed_by` INTEGER REFERENCES `users(id)`
- `server/src/routes/images.js` — PATCH endpoint thêm xử lý `completed_at/completed_by`
- `client/src/types.ts` — thêm `completed_at, completed_by` vào `ImageItem`
- `client/src/pages/AnnotatorPage.tsx` — thêm nút/phím tắt "confirm done"

Phase 4.1 — Detect cache

Files bị ảnh hưởng:

- `server/src/db.js` hoặc file migration riêng — CREATE TABLE `detect_cache`
- `server/src/routes/autolabel.js` — check cache trước khi gọi inference service

Phase 5.1 — Undo/Redo

Files bị ảnh hưởng:

- `client/src/pages/AnnotatorPage.tsx` — thêm `undoStack/redoStack` state, `Ctrl+Z/Ctrl+Y` handler

Phase 6.1 — Model versioning

Files bị ảnh hưởng:

- `server/src/db.js` — ALTER TABLE `models` ADD COLUMN `metadata` JSON
- `server/src/routes/models.js` — thêm PATCH endpoint, GET endpoint với `metadata`
- `client/src/types.ts` — mở rộng `ModelInfo`

10. Lịch sử cập nhật

Ngày	Người cập nhật	Nội dung	Commit
---	---	---	---
2026-08-04	senior-developer (STEP-0.1)	Tạo mới — full audit 5 bảng, 8 routes, API endpoints, blast radius map	(STEP-0.1)
2026-08-04	senior-developer (STEP-1.1)	Cập nhật §3.10 autolabel.js (HTTP mode + legacy rollback), thêm §5.2 inference_service.py, cập nhật §8 Python deps, §9 Phase 1.1 DONE	(STEP-1.1)
2026-08-04	senior-developer (STEP-1.2)	Thêm §3.11 jobs.js (MỚI), cập nhật §2 (migrations/ + jobs.js), §3.1 (route /api/jobs), §3.2 (startup	(STEP-1.2)

		cleanup), §3.10 (DB-backed job storage), §6 (jobs table schema), §7 (/api/jobs endpoints), §9 Phase 1.2 DONE	
2026-08-04	junior-developer (STEP-1.3)	Thêm §3.12 thumbnails.js (MỚI), cập nhật §1 (data/thumbnails/), §2 (thư mục + route), §3.1 (mount /api/images), §3.5 (thumbnail_url field), §4.3 (ImageItem.thumbnail_url), §7 (/api/images/:id/thumb), §9 Phase 1.3 DONE	(STEP-1.3)
2026-08-04	tech-lead+cto+em (STEP-2.1)	Auth ADR riêng (ADR-auth-labeling-studio.md), security-audit-stride, CTO+EM APPROVED kèm điều kiện #A1/#A2	(STEP-2.1)
2026-08-04	senior-developer (STEP-2.2)	Bảng users (INTEGER id, role, color), routes auth.js/users.js, middleware, LoginPage — 87 test pass ma trận AD-A5, npm audit sạch (điều kiện #A1/#A2 đạt), §9 Phase 2.2 DONE	(STEP-2.2)
2026-08-04	senior-developer (STEP-2.3)	Thêm §3.13 reviews.js (MỚI), cập nhật §6 bảng images (4 cột review), §9 Phase 2.3 DONE — review workflow submit/approve/reject, Row 8 test pass	(STEP-2.3)
2026-08-05	senior-developer (STEP-3.1)	Cập nhật §6 — cột annotations.version, bảng mới annotation_history (snapshot), activity_log; §9 Phase 3.1 DONE — verify row count trước/sau (0 mất dữ liệu, CTO condition #1 đạt)	(STEP-3.1)